06 · Is spend *causing* sales? — causal MMM (pymc-marketing)

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> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. Brand search converts like crazy — should we pour budget into it? Be careful: brand search mostly **catches** demand that upper-funnel channels (TV) and seasonality already created, so crediting it for those conversions and shifting budget its way would be a mistake.

1.0.1 The concepts

- **MMM (Marketing Mix Model)** — a regression of sales on each channel's spend over time (usually weekly), used to attribute sales to channels and set budgets. It's the "top-down" alternative to click-tracking.
- **Adstock (carry-over)** — advertising doesn't only work the week it runs; its effect *decays over subsequent weeks*. Adstock transforms raw spend into an "effective" exposure that carries over.
- **Saturation (diminishing returns)** — the 10th €1,000 into a channel buys less lift than the 1st. A saturation curve bends over as spend rises; its *slope* is the **marginal** return that should drive the next euro of budget.
- **Last-click attribution** — crediting the conversion to the final touch (often brand search). It systematically **over-credits** demand-capture channels, which is the error a causal MMM aims to fix.
- **Incremental vs mediated** — the *incremental* contribution of a channel is what it *caused*; a channel can show a big correlation with sales while causing very little (it was downstream of the real driver).

1.0.2 The arc of this notebook — a cautionary tale with a resolution

A plain MMM regresses sales on all channels and is fooled by a **confounder** like seasonality (it lifts sales *and* drives brand-search spend). The textbook fix is a **causal MMM**: hand it a **DAG** and let the backdoor logic (nb 05) control for the confounder. This notebook shows that fix is **necessary but not sufficient**, in three acts:

1. **The confounding correction works — in a linear model.** A naive OLS attribution that omits seasonality massively over-credits brand_search; adding seasonality as a control collapses that credit toward the small truth. That's the cleanest way to *see* the confounding, and it's robust.

2. **The out-of-the-box causal MMM still inverts the ranking.** Fit with library-default priors, the MMM's *flexible saturation curve* for brand_search **re-absorbs** the seasonal demand the linear control was meant to remove — so it credits brand_search *more* than TV, the opposite of the truth. A DAG plus a linear control does **not** de-confound a channel whose spend is collinear with the confounder.
3. **Experiment-calibrated priors recover it.** Feed in what a small **geo experiment** teaches — that brand_search's incremental effect is small — as an informative prior on its saturation, and the MMM ranks the channels correctly again. **This is why MMMs must be calibrated with experiments** (Anchor B, nb 07): observational data alone cannot pin the decomposition.

   On real data. Swap in your **own weekly spend-by-channel + sales** table plus known confounders (seasonality, price, distribution). Public example: the datasets shipped with Meta's *Robyn* or `pymc-marketing`'s own tutorials. Always pair an MMM with occasional geo experiments to anchor the levels.

It follows the repo's fixed **7-step contract**: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

1.1 2 · Simulate a ground truth

Weekly sales from a faithful **adstock + Michaelis-Menten saturation** process. **TV** is a big genuine driver. **seasonality** is a **confounder** — it lifts sales directly *and* drives brand-search spend. **brand_search** has only a *small* real effect, but its spend rides the seasonal wave (corr ~ 0.75), so a naive attribution ignoring seasonality massively over-credits it. Honest answer: TV highly incremental, brand_search only slightly.

```
TRUE incremental sales - TV 5,263   brand_search 518 (small)

   date_week         tv  brand_search  seasonality       sales
0 2023-01-02  42.870187     35.579322     0.000000  134.900978
1 2023-01-09  53.049712     17.732754     0.120537  137.863335
2 2023-01-16  27.623875     23.486826     0.239316  142.263680
3 2023-01-23  56.329747     34.885672     0.354605  159.788875
4 2023-01-30  43.339248     33.736743     0.464723  152.485792
```

1.2 3 · Identify — the marketing DAG picks controls vs mediators

DAG: `seasonality -> {sales, brand_search}`, `tv -> sales`, `brand_search -> sales`. The backdoor criterion says **seasonality is a confounder -> control for it**; omitting it is exactly the last-click mistake that over-credits brand search. The estimand for channel c is its **incremental contribution** — sum over weeks of $\mathbb{E}[\text{sales} \mid do(\text{spend}_c = \text{observed})] - \mathbb{E}[\text{sales} \mid do(\text{spend}_c = 0)]$, holding the other channels at their observed spend (how much of sales would vanish if we switched channel c off). ROI = contribution ÷ spend, each with a posterior. Identification comes from the backdoor set {seasonality}; we hand the DAG to the MMM via `dag`, `treatment_nodes`, `outcome_node`.

```
Backdoor adjustment set the causal layer will use: ['seasonality']
```

1.3 4 · Estimate — fit the causal MMM twice

We fit the **same DAG-aware MMM twice**, to make the lesson unmissable:

- **(a) default priors** — the honest, out-of-the-box causal MMM. Watch it *invert* the ranking.
- **(b) experiment-calibrated priors** — encode what a geo experiment teaches (brand_search's incremental effect is small) as an informative prior on its saturation ceiling, and watch the ranking snap back.

A caution that runs through both: **MMMs are miscalibrated in *absolute* terms** — with channel spend riding seasonal demand, a short weekly series can't fully separate "channel drove sales" from "season drove both", so fitted contributions run high. What we're after is the **ranking** and each channel's **ROI verdict**.

```
Output()
```

```
(a) default-prior MMM convergence: max r-hat 1.010 - min ESS 381 - divergences
14
(a) default contribution - TV 10,390   brand_search 36,007
    true: TV 5,263, brand_search 518  ->  the MMM ranks brand_search ABOVE TV
(inverted!).
```

1.3.1 Why the out-of-the-box MMM inverts the ranking

The default MMM credits brand_search *above* TV — backwards. The mechanism is worth seeing, because it's a real trap:

- brand_search's spend is *built* to ride the seasonal demand wave (Step 2 constructs it from the same seasonal index that drives sales), and the fit lets its flexible saturation curve quietly **re-absorb the seasonal demand** the linear `seasonality` control was meant to remove.
- Because brand_search rides the seasonal wave more tightly than TV does, it wins that tug-of-war and books a huge contribution (tens of thousands vs a true ~500), while a large **negative baseline** offsets it — the classic non-identified MMM level decomposition.
- The DAG was right and the control was present; they still weren't enough. **A linear confounder-control does not de-confound a channel whose spend is collinear with the confounder and enters through a flexible functional form.** Observational data alone can't break the tie.

1.3.2 The fix — priors calibrated from a geo experiment

The standard remedy is exactly the discipline of Anchor B (nb 07): run a **small geo experiment**, learn that brand_search's *incremental* effect is small, and feed that back as an **informative prior on its saturation ceiling** (`saturation_alpha`). TV, whose incremental role we haven't independently pinned down, keeps a weakly-informative prior. This is not putting a thumb on the scale — it's using experimental evidence exactly where observational data is silent.

```
Output()
```

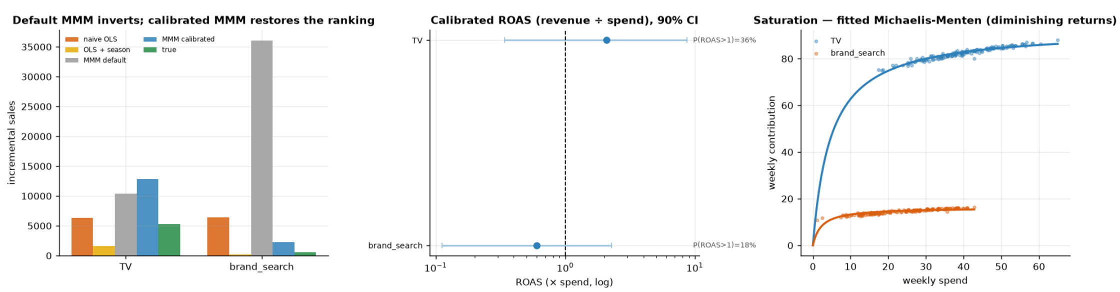
```
(b) calibrated-prior MMM convergence: max r-hat 1.010 - min ESS 376 -
divergences 0
(b) calibrated contribution - TV 12,820   brand_search 2,262
    true: TV 5,263, brand_search 518  ->  ranking RESTORED: TV above
brand_search.
```

1.4 5 · Validate — the correction, the recovered ROI, and saturation

Three views:

1. **The confounding correction, several ways** — naive OLS (omit season) over-credits brand_search; OLS + **season** collapses it toward truth; the **default** causal MMM *inverts* it; the **calibrated** MMM restores the right ranking. The contrast between the two MMMs is the headline.
2. **ROI with uncertainty (calibrated MMM)** — each channel's posterior ROI vs the break-even line. ROAS = contribution ÷ (fixed **observed** spend), so it carries the **same relative miscalibration as the contribution** — dividing by a constant spend cannot remove the bias. What is robust is the ROAS **ranking** (and brand_search's below-break-even verdict), **not** the absolute ROAS *levels*, which are biased by the same factor as the contributions.
3. **Saturation curves** — contribution vs spend per channel, showing diminishing returns (the shape that sets *marginal* ROI and hence where the next euro should go).

```
brand_search credit - naive 6,401 | OLS+season 222 | MMM default 36,007 | MMM
calibrated 2,262 | true 518
calibrated ROAS - TV 2.08x [90% 0.34,8.67]  ·  brand_search 0.60x [90%
0.11,2.28]
TRUE ROAS       - TV 0.85x · brand_search 0.14x  (both below 1x: the MMM
recovers the ranking but overstates levels)
```



How to read this. *Left* tells the whole story in one panel. **Naive OLS** (orange) over-credits brand_search; **OLS + season** (gold) collapses that credit toward the small truth — the confounding correction, working. Then the two MMMs: the **default** causal MMM (grey, the failure mode) *inverts* the ranking, crediting brand_search far above TV — the seasonal-demand-stealing trap — while the **calibrated** MMM (blue) restores TV above brand_search, landing beside the **true** bars (green). *Middle* — the decision-relevant **calibrated ROI**: TV's mean clears break-even but its 90% interval is wide and dips below 1x, while brand_search's mean sits **below break-even**. So TV is the better channel, but its above-break-even verdict is itself uncertain. And the ranking is only *probably* right: draw-by-draw, TV's ROI beats brand_search's about **78%** of the time in this run (the `P(TV ROI > brand_search ROI)` computed next) — more likely than not, but **short of the 0.8 bar** we'd want before moving real budget. *Right* — the **saturation curves**: contribution flattens as spend rises (diminishing returns), so the *next* euro into a channel is worth its curve's *slope*, not its average. The honest caveat (Step 7): even the calibrated ROI *levels* are approximate — anchor them with the same kind of geo experiment that supplied the prior.

1.5 6 · Decide, in euros — should we reallocate budget yet?

Budget chases **incremental return**, not last-click credit. One naming point first: what the MMM gives us is **ROAS** — incremental *revenue* per euro of spend — not profit ROI. Break-even at 1x therefore assumes a 100% margin; at a realistic margin *m* the break-even ROAS is **1/m** (a 30% margin -> 3.3x), which we show below.

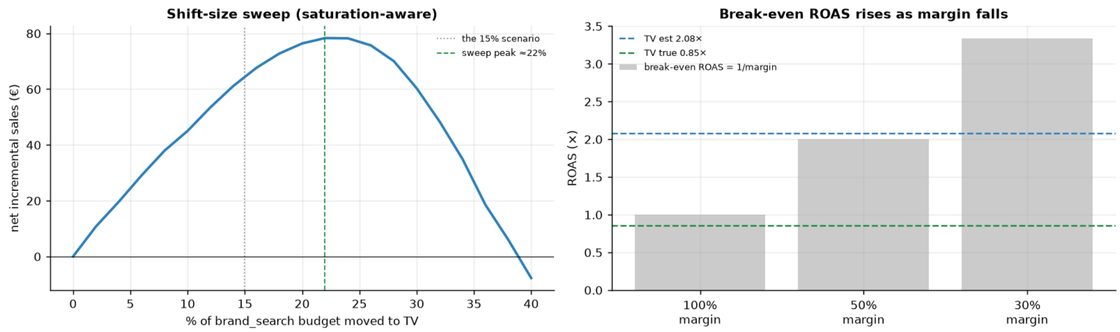
Now that experiment-calibrated priors have made the *ranking* trustworthy, the point estimates put TV above brand_search, so the *direction* of any reallocation is toward TV. But a direction is not a decision, and the **absolute levels are still miscalibrated**. So we quantify three things: (1) the sales impact of shifting a slice of budget — swept across shift sizes and made **saturation-aware**, so marginal ROAS falls as we pour more into TV; (2) the posterior probability the move actually helps; and (3) the honest gap between the MMM's ROAS and the *true* ROAS. That last check is the sobering one: it is why, even with the ranking fixed, the absolute "is TV worth it?" question still belongs to a geo test (Anchor B, nb 07), not a still-approximate MMM.

```
TV ROAS - average 2.08x · marginal (the next euro, at today's spend) 0.23x -
marginal is lower because of saturation (diminishing returns).

Shifting 15% of brand_search spend (€566) to TV -> +€61 net sales (saturation-
aware); the sweep peaks near 22%.
P(TV ROAS > brand_search ROAS) = 0.78 -> lean toward TV, but test first

Ranking vs levels - the watertight part:
  TRUE ROAS: TV 0.85x · brand_search 0.14x (both BELOW 1x)  |  MMM says TV 2.08x
· bs 0.60x
  The MMM recovers the RANKING (TV >> brand_search) but overstates LEVELS ~2.4x
for TV.
  ROAS is revenue ÷ spend - it assumes 100% margin. At a realistic 30% margin
the break-even is 1/0.30 ~ 3.3x,
  which even TV's estimated 2.08x fails; and TV's TRUE ROAS 0.85x is below
break-even at ANY margin.
  So: reallocate toward TV (the ranking is trustworthy), but 'is TV worth it in
absolute €?' needs the geo
  experiment (nb 07) - this MMM's absolute levels are not calibrated enough to
answer it.

Decision quantity WITH uncertainty - shift 15% (€566) brand_search -> TV (per-
draw, average-ROAS):
  Deltasales = €836  [90% €-799, €4,618]  ·  P(Delta>0) = 0.78
  TRUE Deltasales (from DGP true_contrib) = €405  ->  direction check: MATCH
(both favour TV)
```



Which Deltasales number goes to the CFO? Two figures print above for the same 15% shift, and they differ ~13x on purpose: **+€61/week** is the *saturation-aware* estimate — it walks both channels down their fitted response curves, and is the honest planning number; **+€836/week** prices the shift at *average* ROAS per posterior draw — it ignores saturation entirely and is kept only because it carries the uncertainty band (its P(Delta>0)~0.78 is the ranking probability in euro clothing — by construction Delta>0 exactly when TV's ROAS beats brand_search's). Quote €61; use €836's band as the direction check it is.

1.6 7 · Caveats

- **Absolute contributions need calibration.** Even the calibrated MMM runs a fewx high in absolute euros, because channel spend rides the seasonal wave the model can't fully net out from a short series. **Trust the ranking and the ROI verdict, not the absolute euros** — and note it took an experiment-informed prior to get even the *ranking* right. This is the single most important thing to say about any MMM.
- **A DAG is necessary but not sufficient.** The correct backdoor control (seasonality) did not stop a flexibly-saturated, collinear channel from re-absorbing the confounder. Structural assumptions still need experimental calibration to be trustworthy.
- **Adstock/saturation priors matter — a lot.** Carry-over (lambda) and diminishing-returns (alpha, kappa) are weakly identified from limited weekly data; here the saturation-ceiling prior *flipped the conclusion*. Use informative priors from experiments wherever you have them.
- **Correlated channels -> wide, correlated ROI posteriors.** When spend moves with a confounder, the data can't fully separate them; the intervals will (honestly) show it.